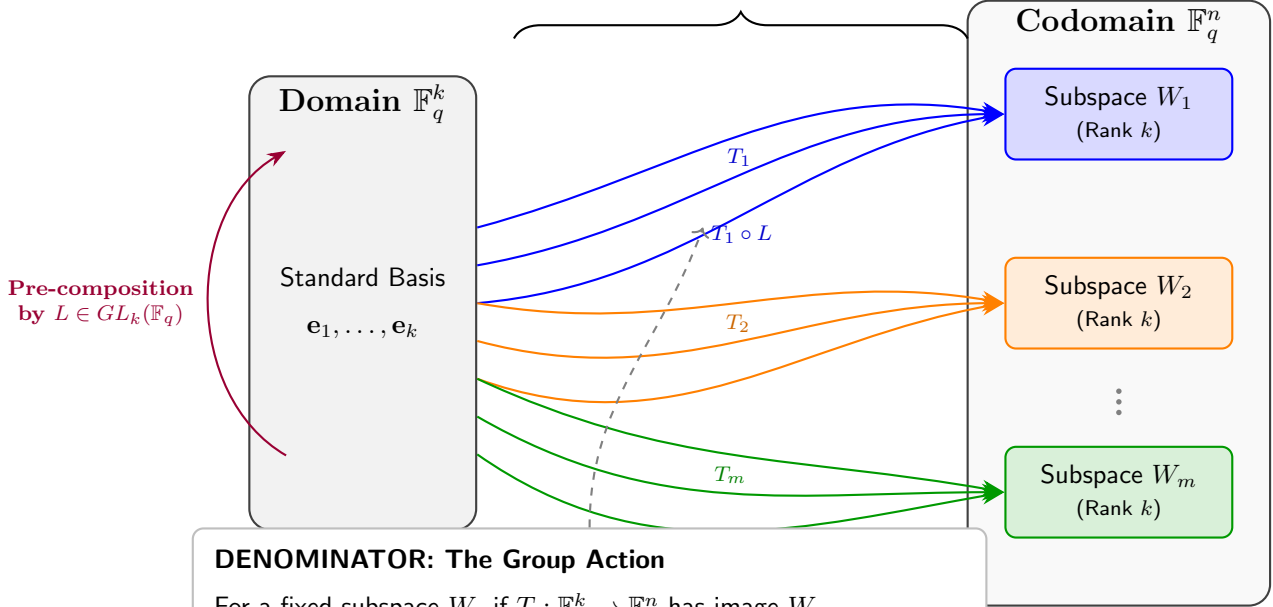


NUMERATOR: All Injective Linear Operators $T : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$

Total size: $(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})$



DENOMINATOR: The Group Action

For a fixed subspace W , if $T : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$ has image W , then every other operator mapping to W is of the form $T \circ L$, where $L : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^k$ is a bijective linear operator.

Size of equivalence class $= |GL_k(\mathbb{F}_q)| = (q^k - 1) \cdots (q^k - q^{k-1})$

$$\text{Distinct Subspaces} = \frac{|\text{Injective Operators } \mathbb{F}_q^k \rightarrow \mathbb{F}_q^n|}{|\text{Automorphisms } GL_k(\mathbb{F}_q)|}$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}$$